VANTARA

Smart Supply Chain Intelligence Platform

Business Requirements & Analysis Document

SQL Business Question Analysis — Findings & Recommendations

1. Executive Summary

This document translates ten SQL window-function analyses performed on Vantara's supply chain dataset (fact_orders, dim_customers, dim_products, dim_location, dim_date) into business-facing findings, impact statements, and recommendations. Each question was chosen because it maps directly to a KPI or visual already live on the Power BI report (Overview, Delivery Performance, Customer Insights pages).

Across the ten queries, three issues stand out as needing action before the next stakeholder review: a geographic scope mismatch on the state-level revenue map, a data grain inconsistency on delivery status counts, and an unrealized opportunity to add customer-tier and order-cadence views that the underlying data already supports.

2. Business Question Analysis

Q1. Who are our highest-value customers by lifetime spend?

Finding: Ranking customers by total sales shows the top customer has generated $10,524.17 in lifetime revenue, with a long tail beneath the top 10 customers cluster between $9,000 and $10,500.

Business Impact: A small group of customers is disproportionately valuable. Losing even one of these accounts has outsized revenue impact compared to losing an average customer.

Recommendation: Stand up a VIP/high-value customer segment in the CRM and route it through a dedicated retention or account-management workflow. Feed this ranking into the Customer Insights page as a top-customers table.

Priority: Medium

Q2. How is cumulative revenue trending month over month?

Finding: Running-total revenue climbs steadily through the year, consistent with the monthly bars already shown on the Overview page's Revenue Trend chart.

Business Impact: Leadership can see year-to-date pace at a glance instead of reading individual monthly bars, which is useful for tracking progress against annual targets.

Recommendation: Add a running-total line overlay on the existing Revenue Trend chart so cumulative pace and monthly seasonality are visible in one view.

Priority: Low

Q3. Which months saw the sharpest revenue swings, and why?

Finding: Month-over-month growth is volatile for example, a -11.85% drop in February 2015 followed by a +13.40% rebound in March. This is masked by the single 47.91% Sales Growth Rate card on the Overview page, which does not indicate whether that figure is a period average, a single comparison, or something else.

Business Impact: A flat headline growth number can hide meaningful volatility. Decision-makers reading only the card may misjudge how stable the business actually is month to month.

Recommendation: Clarify the DAX logic behind the Sales Growth Rate card (confirm the exact period it compares) and consider adding a small trend sparkline or footnote so the card's number isn't read as a constant.

Priority: High

Q4. Which product categories are most profitable, and does this differ by customer segment?

Finding: Fishing, Cleats, Camping & Hiking, Cardio Equipment, and Women's Apparel are the top five profitable categories, and notably this ranking holds in the same order for both Consumer and Corporate segments.

Business Impact: Category strategy does not need to be customized per segment; the same assortment and merchandising priorities work across customer types, simplifying inventory and marketing planning.

Recommendation: Prioritize inventory investment and promotional spend on the top two categories (Fishing, Cleats) platform-wide rather than building segment-specific category strategies.

Priority: Low

Q5. How do we correctly identify repeat vs. one-time customers?

Finding: Numbering each customer's orders chronologically and flagging any customer with more than one order confirms the repeat-customer logic used in the Power BI Repeat Customer Rate card and donut chart (56.98% repeat) is built on the correct definition.

Business Impact: Repeat rate is a headline retention metric reported to leadership. An incorrect definition would misstate business health and could lead to wrong retention-strategy decisions.

Recommendation: Lock this order-sequence logic in as the single source of truth for 'repeat customer' across all reporting (SQL and Power BI), and document it so future measures don't drift from this definition.

Priority: High

Q6. Which states/regions drive the most revenue?

Finding: England ('Inglaterra'), Isle of France, and California lead by revenue share, but the underlying location dimension includes international markets while the Power BI map visual only renders U.S. states leaving most U.S. states gray and effectively hiding non-U.S. revenue entirely.

Business Impact: The current map understates the platform's international footprint and may lead stakeholders to believe the business is more U.S.-concentrated than it actually is.

Recommendation: Replace the U.S.-only state map with a world map visual, or add a market filter/toggle so viewers can switch between domestic and global views.

Priority: High

Q7. Is delivery/fulfilment time stable or trending in a concerning direction?

Finding: A 3-month moving average of fulfilment time smooths out month-to-month noise and shows fulfilment hovering close to 3.5 days across early 2015, consistent with the Delivery Performance page's 3.47-day average card.

Business Impact: Smoothed trend data is more reliable for spotting genuine service-level drift versus one-off delays, which matters for setting realistic customer delivery-time expectations.

Recommendation: Add a fulfilment-time trend line (using the 3-month moving average) to the Delivery Performance page so operations can catch gradual slippage before it shows up as a customer complaint spike.

Priority: Medium

Q8. Can we segment customers into value tiers for targeted marketing?

Finding: Splitting customers into four spend-based quartiles is straightforward with the existing data (top quartile customers spend $8,595–$10,524 lifetime), but no such tiering currently exists anywhere in the Power BI report.

Business Impact: Marketing and retention teams currently have no data-driven way to prioritize outreach; all customers are treated the same regardless of value.

Recommendation: Build a customer-tier table or visual on the Customer Insights page using this quartile logic, and use it to drive differentiated marketing (e.g., loyalty perks for Tier 1, win-back offers for Tier 4).

Priority: Medium

Q9. How frequently do customers reorder, and is that changing?

Finding: Measuring days between a customer's consecutive orders shows wide variation one customer's second order came 273 days after their first, then the third came only 144 days after that, suggesting purchase cadence can accelerate after the first repeat.

Business Impact: Understanding reorder cadence lets the business time win-back campaigns (e.g., trigger an email if a customer passes their typical reorder window without a new order).

Recommendation: Add an order-cadence or 'days since last order' distribution to Customer Insights, and use it to power a lifecycle/win-back email trigger.

Priority: Medium

Q10. What is the biggest driver of delivery problems?

Finding: Late delivery accounts for 54.83% of all order-line records (98,977 of ~180,519), far ahead of shipping cancellations at 4.30%. This confirms the report's own narrative text but the underlying count doesn't match the 36.0K late-delivery figure shown in the Delivery Performance bar chart pointing to a difference between counting order lines versus distinct orders.

Business Impact: Late delivery is unambiguously the top operational risk, but the mismatched counts mean the report may currently understate (or misrepresent) the true scale of the problem depending on which grain stakeholders assume they're reading.

Recommendation: Reconcile the SQL and DAX so both count at the same grain (order-line vs. distinct order_id), then treat late delivery as the top operations priority investigate root cause by shipping mode and carrier.

Priority: High

3. Consolidated Recommendations

The following actions are grouped by priority based on business risk and reporting accuracy impact.

High Priority — Fix Before Next Review

- Clarify and document the exact calculation behind the Sales Growth Rate card so it isn't misread as a constant figure (Q3).
- Lock the repeat-customer definition (order sequence > 1) as the single source of truth across SQL and Power BI (Q5).

- Fix the state revenue map's geographic scope — switch to a world map or add a market filter so international revenue isn't hidden (Q6).
- Reconcile order-line vs. distinct-order counting so delivery status figures match between SQL and the dashboard (Q10).

Medium Priority — Adds Analytical Depth

- Introduce a high-value/VIP customer segment for retention workflows (Q1).
- Add a fulfilment-time trend line using the 3-month moving average (Q7).
- Build a customer value-tier (quartile) view to support targeted marketing (Q8).
- Add an order-cadence view to support win-back campaign triggers (Q9).

Low Priority — Nice to Have

- Overlay cumulative running-total revenue on the existing trend chart (Q2).
- No action needed on category profitability — current ranking is stable across segments and already reflected accurately (Q4).